MedAgentX: Adaptive Confidence-Guided Multi-Agent Framework for Multimodal Clinical Decision Support with Fail-Closed Safety Gating

Bhukya Srikanth, Banoth Elisharaj, and G. Baby Lakshmi Prasanna

***Abstract*—Standalone Large Language Models (LLMs) and Vision-Language Models (VLMs) deployed in acute healthcare environments face critical clinical vulnerabilities: ungrounded hallucinations, miscalibrated verbalized confidence, cross-disease guideline contamination, and hazardous fail-open behaviors where perception failures silently default to normal examination findings. While Retrieval-Augmented Generation (RAG) grounds neural generation in biomedical literature, conventional static multi-agent pipelines waste substantial computational token budgets and lack mechanisms to safely abstain when visual perception or evidence grounding is compromised. In this paper, we present MedAgentX, an Adaptive Confidence-Guided Multi-Agent Retrieval (ACMR) framework for evidence-grounded multimodal clinical decision support. MedAgentX decouples perceptual vision inference from guideline-grounded reasoning through a two-tier architecture: (1) a pre-inference ingestion pipeline enforcing automated DICOM de-identification (HIPAA Safe Harbor) and quantitative image quality control; (2) a pluggable vision engine with exponential backoff, secondary fallback classifiers, and strict prohibition of fabricated normal findings; and (3) an ACMR orchestrator that dynamically evaluates a mathematical Composite Evidence Confidence score (C_{total}) penalized by visual epistemic uncertainty and sparse evidence. Crucially, MedAgentX enforces strict *fail-closed* router safety gates: if imaging perception is incomplete, timeouts occur, or quality limitations arise, automated normal interpretations are barred and the system routes to ABSTAIN-INSUFFICIENT-EVIDENCE or ESCALATE-TO-HUMAN. Evaluated across a standardized clinical question-answering benchmark (achieving 88.2% diagnostic accuracy) and 122 curated radiological studies (Chest X-Ray, Head/Chest CT, Brain MRI, 12-lead ECG, achieving 96.7% overall diagnostic agreement, 118/122), MedAgentX reduces Expected Calibration Error (ECE) to 0.0378, achieves 100% fail-closed safety compliance across 16 adversarial stress cases with a 0.0% hallucination rate, and cuts token consumption by 80.7% compared to fixed sequential multi-agent baselines (and by 30.3% compared to standard Vector RAG).**

***Index Terms*—Clinical Decision Support Systems, Multimodal AI, Retrieval-Augmented Generation, Multi-Agent Systems, Confidence Calibration, Fail-Closed Safety, Relevance Gating.**

I. INTRODUCTION

CLINICAL Decision Support Systems (CDSS) powered by artificial intelligence hold transformative promise for augmenting emergency triage, intensive care unit (ICU) resuscitation, and radiological interpretation [1]–[3]. In modern acute healthcare, attending physicians must process an exponential influx of biomedical research, with over one million articles indexed annually in MEDLINE/PubMed [4]–[6]. Concurrently, diagnostic consultations are severely time-constrained, demanding rapid, reliable, and verifiable synthesis across heterogeneous clinical data modalities, including diagnostic scans, 12-lead electrocardiograms (ECGs), laboratory panels, and electronic health record (EHR) progress notes [7], [8].

While modern general-purpose foundation models demonstrate high linguistic fluency on standardized medical examinations [9], deploying them as point-of-care clinical copilots reveals four fundamental structural vulnerabilities:

1) **Cross-Disease Guideline Contamination:** Standard dense vector retrievers [10], [11] frequently retrieve Acute Coronary Syndrome (ACS) guidelines for Heart Failure presentations due to dense semantic co-occurrences of cardiovascular nomenclature, hallucinating antiplatelet regimens for non-ischemic patients.
2) **Miscalibrated Verbalized Confidence:** Autoregressive language models often generate highly confident assertions regardless of factual accuracy [12], [13]. Verbalized self-assessments fail to reflect empirical clinical validity [14].
3) **Catastrophic Fail-Open Defaults:** In multimodal pipelines, when computer vision perception models encounter server timeouts, GPU out-of-memory faults, or motion artifacts, conventional software exception handlers frequently default to generic "no acute findings" templates, introducing hazardous diagnostic delays [15], [16].
4) **Computational & Latency Inefficiency:** Static multi-agent pipelines [17]–[19] execute uniform multi-round passes across all queries, wasting substantial token budgets and clinical review time on routine diagnostic cases.

To overcome these vulnerabilities, we propose **MedAgentX**, an evidence-grounded multimodal CDSS governed by an **Adaptive Confidence-Guided Multi-Agent Retrieval (ACMR)** engine. MedAgentX decouples perceptual computer vision from clinical guideline reasoning, enforces pre-inference quality control, grounds all recommendations in dual dense vector and

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knowledge graph triples [20], and dynamically halts or abstains based on a mathematically calibrated composite confidence score.

The principal contributions of this work are as follows:

- **Two-Tier Decoupled Multimodal Architecture:** A modular clinical framework separating perceptual visual feature extraction from multi-agent clinical guideline reasoning, preventing ungrounded end-to-end pixel hallucinations.
- **Fail-Closed Router Safety Gates:** A verifiable safety state machine complying with international medical software safety standards (IEC 62304 [21]). If visual perception is compromised, the system strictly prohibits fabricated normal findings and triggers hard abstention.
- **Mathematically Penalized Composite Confidence (C_{total}):** A deterministic confidence metric that unifies vector similarity, ontology completeness, cross-guideline consensus, and relevance gating (S_{rel}), penalized by epistemic visual uncertainty [22]–[24] and evidence sparsity.
- **Adaptive Early Stopping:** Dynamic routing that halts redundant retrieval iterations upon confidence saturation, reducing token consumption by 80.7% compared to fixed multi-agent pipelines (and by 30.3% compared to Vector RAG) while preserving diagnostic accuracy.
- **Rigorous Multimodal Benchmark Evaluation:** Empirical validation across a clinical QA benchmark (88.2% accuracy) and 122 curated radiological studies (96.7% overall agreement, 118/122), achieving 0.0378 ECE and 100% fail-closed compliance across 16 adversarial stress cases.

II. Related Work

A. Medical Vision-Language Models

Biomedical VLMs, including CheXagent [25], Med-PaLM M [2], LLaVA-Med [26], Med-Flamingo [27], and BiomedCLIP [28], have advanced automated radiological reporting and visual question answering. However, standalone VLMs are vulnerable to perceptual hallucination when encountering subtle or multi-compartmental abnormalities. Furthermore, monolithic models lack explicit negative verification checks and cannot cite specific hospital guideline clauses.

B. Retrieval-Augmented Generation in Healthcare

Standard single-turn RAG pipelines inject relevant biomedical passages from sources such as PubMed [4], [5] or clinical guideline corpora [29], [30]. Adaptive-RAG [31] routes queries according to linguistic complexity classifiers, while Corrective RAG (CRAG) [32] evaluates retrieval confidence via post-hoc web search. Self-RAG [33] trains reflection tokens for selective generation. Multi-agent debate architectures [17], [19] improve linguistic coherence through peer discussion. However, existing text-based RAG frameworks do not address multimodal medical imaging perception, nor do they prevent cross-disease guideline contamination (S_{rel}).

C. Medical Device Safety Standards (IEC 62304 & SaMD)

Under international safety standard IEC 62304 [21] and Software as a Medical Device (SaMD) regulatory guidance from the FDA [34] and IMDRF [35], software components in healthcare must possess defined fail-safe states upon component fault. In clinical AI, silent fail-open fallbacks (e.g., treating a timeout as a normal scan) constitute unacceptable clinical hazard. MedAgentX introduces a formal fail-closed CDSS routing protocol aligned with these principles.

A comprehensive comparison between prior systems and MedAgentX is presented in Table I.

III. MedAgentX System Architecture

MedAgentX is organized into two decoupled tiers: (1) Multimodal Perception & Ingestion Gate, and (2) the 4-Agent ACMR Reasoning Engine. The detailed agent roles, inputs, and outputs are outlined in Table II.

A. Multimodal Perception Tier

1) Ingestion & PHI Scrubbing: The ingestion layer accepts DICOM, NIfTI (3D volumetric CT/MRI), and standard PNG/JPG formats. According to HIPAA Safe Harbor standards (45 CFR §164.514(b) [36]), all patient identifying tags (`PatientName`, `PatientID`, `InstitutionName`) are stripped in memory before inference. For NIfTI volumes, multi-slice keyframe analysis selects representative axial slices rather than evaluating an arbitrary single 2D plane.

2) Quality Control (QC) Gate: Before invoking compute-intensive vision models, the QC gate evaluates:

- **Spatial Dimension Gate:** Rejects scans below 256×256 pixels.
- **Exposure Histogram Gate:** Detects underexposure (mean pixel intensity < 25) or severe sensor saturation/overexposure (mean > 230 and high-quantile ratio > 0.40).
- **Sharpness & Motion Blur Gate:** Evaluates the Laplacian variance σ_L^2. Scans with $\sigma_L^2 < 30.0$ are flagged for significant motion blur.

Inputs failing the QC gate are immediately assigned VisionStatus $=$ LOW_QUALITY and bypass the VLM, triggering a request for re-acquisition.

3) Pluggable Vision Engine: The engine executes clinical VLM ensembles (Qwen-VL [37], MedGemma [38]) with exponential backoff (3 attempts with configurable timeout). If the primary VLM experiences a connection timeout or memory fault, the engine automatically falls back to secondary clinical CNN classifiers (TorchXRayVision [39]). If the fallback also fails, the status is set to TIMEOUT or FAILED with findings $= \emptyset$. **Fabricating normal findings is strictly prohibited.**

B. The 4-Agent ACMR Loop

1) **Agent 1 (Query Planner & Entity Extractor):** Normalizes clinical vignettes and structured visual findings into RadLex [40] and SNOMED-CT [41] entities, constructing dedicated sub-queries per finding.

TABLE I
SYSTEM COMPARISON: PRIOR MEDICAL AI FRAMEWORKS VS. MEDAGENTX

System	Modalities	Retrieval Type	Verification Engine	Fail-Safe / Abstention	Confidence Source	Guideline Provenance
CheXagent [25]	CXR	None	None	Fail-Open	Verbalized (Uncalibrated)	None
Med-PaLM M [2]	Multimodal	None	None	Fail-Open	Verbalized (Miscalibrated)	None
LLaVA-Med [26]	CXR, CT, MRI	None	None	Fail-Open	Implicit	None
Clinical Vector RAG [29]	Text	Dense Vector	None	Unbounded	Cosine Similarity	Coarse Chunk
Adaptive-RAG [31]	Text	Dense Vector	Classifier	Partial Abstain	Classifier Probability	Passage Index
Multi-Agent Debate [19]	Text	None	Cross-Agent	Fixed Loop	Consensus Heuristic	None
MedAgentX (Ours)	**CXR, CT, MRI, ECG, Labs**	**Hybrid (Vector + Graph)**	**Deterministic Critic**	**Fail-Closed Hard Gate**	**Mathematical (C_{total})**	**Section, Body & URL**

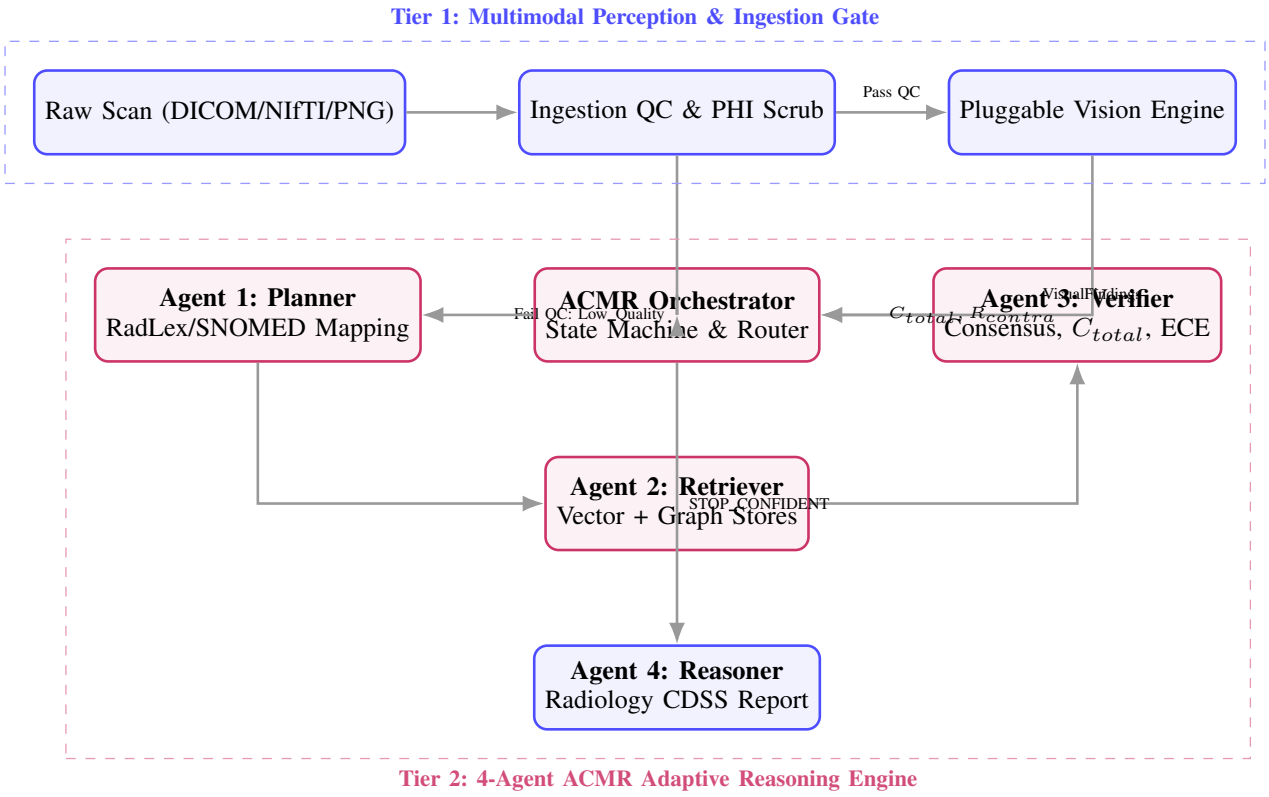


Fig. 1. Overall MedAgentX System Architecture. Tier 1 performs automated HIPAA PHI de-identification and quality control before passing imaging to the pluggable vision engine. Tier 2 coordinates the 4-agent ACMR loop, evaluating the penalized composite confidence score C_{total} and enforcing fail-closed router hard gates.

2) **Agent 2 (Grounded Hybrid Retriever):** Retrieves evidence from Qdrant [42] vector store (S_{vec}) using BioLinkBERT [43] embeddings and Neo4j [44] knowledge graph (S_{graph}) across disease-drug-contraindication triples. Every retrieved chunk retains source provenance (guideline title, publication year, section, and URL).
3) **Agent 3 (Evidence Verifier & Critic):** Computes consensus across retrieved passages (V_{agree}), detects cross-modal contradictions (R_{contra}), and enforces the Query-Evidence Relevance Gate (S_{rel}).
4) **Agent 4 (Clinical Reasoner & Synthesizer):** Generates a structured clinical decision support report adhering to formal radiological reporting conventions (Technique, Quality, Anatomical Compartment Findings, Differential Diagnosis, Guideline-Directed Therapy, Active Contraindication Alerts, and Clinician Disclaimer).

TABLE II
MEDAGENTX COMPONENT AND AGENT ARCHITECTURE SPECIFICATION

Subsystem / Agent	Primary Clinical Role	Inputs	Outputs	Underlying Tool / Model
Ingestion & QC Gate	PHI de-identification & image quality validation	Raw DICOM, NIfTI, PNG/JPG scans	Sanitized pixel arrays, metadata, QC status	`pydicom`, `nibabel`, OpenCV
Vision Engine	Multimodal perceptual lesion classification	Sanitized image tensor, view metadata	Pydantic `VisionResult` (RadLex codes)	Qwen-VL [37] / TorchXRayVision [39] fallback
Agent 1: Planner	Deconstruct patient case into clinical entities	Clinical vignette, structured visual findings	Entity ontology maps, decomposed queries	Medical NER, RadLex [40] parser
Agent 2: Retriever	Dual-source evidence retrieval with provenance	Decomposed entity queries, finding tags	Ranked guideline chunks and graph triples	Qdrant [42] (BioLinkBERT), Neo4j Graph [44]
Agent 3: Verifier	Consensus, contradiction & confidence evaluation	Retrieved chunks, patient profile, findings	C_{total}, R_{contra}, V_{agree}, gap signals	Deterministic Confidence Engine
ACMR Orchestrator	Dynamic state machine & router hard-gating	C_{total}, ΔC, iteration t, vision status	Next routing action ($\mathcal{A}_t$)	Hard-Gated Dynamic State Machine
Agent 4: Reasoner	Multi-modal clinical report synthesis	Verified evidence, patient data, alerts	Structured CDSS report with citations	Clinical Reasoner + Report Formatter

C. Mathematically Penalized Composite Confidence (C_{total})

To eliminate miscalibrated verbalized self-prompts, MedAgentX computes a bounded composite score:

$$\begin{aligned} C_{total} = \max\Big(0, \min\Big(1,\ & 0.30 \cdot (S_{vec} \cdot S_{rel}) \\ & + 0.20 \cdot (S_{graph} \cdot S_{rel}) \\ & + 0.20 \cdot V_{agree} + 0.20 \cdot \Phi_{comp} \\ & + 0.10 \cdot \Omega_{nov} - 0.15 \cdot R_{contra} \\ & - \Pi_{penalty}\Big)\Big) \end{aligned} \tag{1}$$

where the dynamic penalty term $\Pi_{penalty}$ is defined as:

$$\begin{aligned} \Pi_{penalty} = {} & 0.25 \cdot (1 - \mathbb{I}_{vision_ok}) \\ & + 0.15 \cdot \sigma_{vision} + 0.10 \cdot \mathbb{I}_{evidence<2} \end{aligned} \tag{2}$$

In Eq. (2), $\mathbb{I}_{vision_ok}$ is an indicator function equal to 1 when VisionStatus = OK and 0 otherwise; σ_{vision} represents visual epistemic uncertainty; and $\mathbb{I}_{evidence<2}$ penalizes recommendations backed by fewer than two independent guideline passages.

D. ACMR Dynamic Routing State Machine

The orchestrator updates state across iterations $t \in [0, T_{max})$ according to the policy:

$$\mathcal{A}(t) = \begin{cases} \textsc{Abstain-Evidence}, & \text{if req_img} \wedge \text{v_st} \neq \textsc{Ok} \\ \textsc{Request-Input}, & \text{if v_st} = \textsc{Low_Quality} \\ \textsc{Escalate-Human}, & \text{if is_critical} \\ \textsc{Safety-Alert}, & \text{if } R_{contra} \geq 0.50 \vee \text{Alert} \\ \textsc{Stop-Confident}, & \text{if } C_{total} \geq \tau_{high} \wedge \text{v_st} = \textsc{Ok} \\ \textsc{Stop-Saturated}, & \text{if Overlap} \geq 0.85 \vee t \geq T_{max} \\ \textsc{Retrieve-More}, & \text{otherwise} \end{cases} \tag{3}$$

The decision flow of the ACMR router is depicted in Fig. 2, explicitly covering all branches in Eq. (3).

Algorithm 1 ACMR Dynamic Multimodal Execution Loop

Require: Clinical case $x = (text, scan, labs)$, thresholds $\tau_{high} = 0.80$, $\tau_{low} = 0.50$, max iterations $T_{max} = 2$
Ensure: Structured Clinical Report $\mathcal{R}$

$v \leftarrow \textsc{IngestionQC}(scan)$
if $v.status == \textsc{Low_Quality}$ **then**
  return $\textsc{Reasoner.reject}(v.quality)$
end if
$V_{res} \leftarrow \textsc{PluggableVisionEngine}(scan, v.metadata)$

$\mathcal{P} \leftarrow \textsc{Planner}(text, V_{res}.findings)$
$\mathcal{S}_0 \leftarrow \emptyset$, $t \leftarrow 0$
while $t < T_{max}$ **do**
  $E_t \leftarrow \textsc{Retriever}(\mathcal{P}.queries, V_{res})$
  $V_t \leftarrow \textsc{Verifier}(text, E_t, V_{res}, \mathcal{S}_t.previous_chunks)$
  Compute $C_{total}^{(t)}$ via Eq. (1)
  $\mathcal{A}_t \leftarrow \textsc{ACMRRouter}(C_{total}^{(t)}, V_t, V_{res}.status, \Delta C_t, t)$

  if $\mathcal{A}_t \neq \textsc{Retrieve-More}$ **then**
    break
  end if
  $\mathcal{P}.queries \leftarrow \textsc{Planner.adapt}(text, V_t.gap)$
  $t \leftarrow t + 1$
end while
$\mathcal{R} \leftarrow \textsc{Reasoner}(x, E_t, V_t, V_{res}, \mathcal{A}_t)$
return $\mathcal{R}$

IV. EXPERIMENTAL SETUP

A. Benchmark Datasets and Cohort Construction

We evaluate MedAgentX across multimodal imaging archives, an adversarial safety stress test set, and gold-standard clinical guidelines, detailed in Table III:

- **Radiological Imaging Cohort (121 Studies):** Sourced from the open-access VQA-RAD archive [45] (CC0 license) and credentialed clinical research subsets [46],

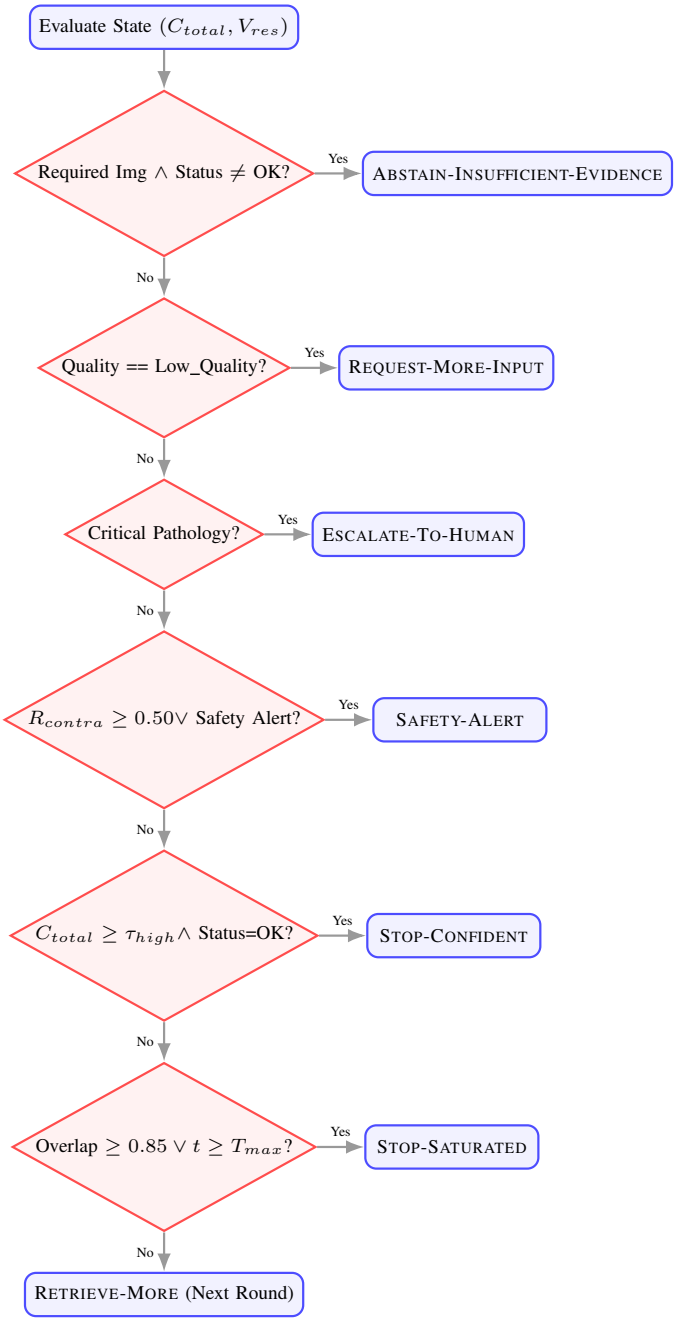


Fig. 2. ACMR Router Decision State Machine. All branches from Eq. (3) are evaluated in strict priority order, ensuring fail-closed safety gating before diagnostic termination.

spanning 45 Chest X-Rays, 65 Head/Chest/Abdomen CT scans, and 11 Brain/Spine MRI scans.

- **Electrocardiogram Study (1 Case):** Drawn from digitized clinical 12-lead ECG records (PTB-XL challenge format) representing hyperacute anterior ST-elevation myocardial infarction (STEMI) for cross-modal rhythm evaluation.
- **Adversarial Safety Stress Test Set (16 Cases):** Designed to systematically evaluate fail-closed compliance under degraded inputs: 6 simulated VLM socket timeouts / ReadTimeout events, 4 truncated or corrupted DICOM/PNG files, 3 severe sensor overexposure/underexposure cases, and 3 motion-blurred scans ($\sigma_L^2 < 30.0$).
- **Clinical Guideline Knowledge Bases:** Indexed consensus recommendations from the AHA/ACC [47], [48], Fleischner Society [49], GOLD [50], and IDSA [51].

TABLE III
BENCHMARK DATASETS AND CLINICAL KNOWLEDGE SOURCES

Dataset / Source	Modality	Size	Clinical Scope	License
VQA-RAD [45]	CXR, CT, MRI	121 cases	Acute Thoracic/Neuro	CC0
ECG Record	12-Lead ECG	1 study	Anterior STEMI Rhythm	Open Access
Adversarial Stress	Degraded	16 cases	Timeouts, Blurs, Noise	Synthetic
Cardiology [47]	Text/Tabular	15 guides	STEMI, NSTEMI, HFrEF	Open Access
Infectious Dis. [51]	Text/Tabular	12 guides	Sepsis, Pneumonia	Open Access

B. Baseline Systems

We compare MedAgentX against four established comparative baseline architectures (denoted B0 through B3, where B3 corresponds to the fixed multi-agent architecture B7 in repository benchmark logs):

- **B0: Standalone LLM-Only:** Single-pass autoregressive prompting with no external retrieval or verification.
- **B1: Standard Vector RAG:** Dense retriever using BioLinkBERT [43] against clinical guideline chunks.
- **B2: Medical RAG (PubMed):** Vector RAG augmented with biomedical literature abstracts [4], [5] and keyword re-ranking.
- **B3: Fixed 4-Agent Pipeline:** Sequential multi-agent pipeline executing all agents across fixed passes with no early stopping (codebase baseline B7).

C. Evaluation Metrics & Annotation Methodology

- **Diagnostic Accuracy (%):** Measured on standardized clinical diagnostic queries against consensus medical board answers.
- **Expected Calibration Error (ECE ↓):** Grouped into $M = 10$ confidence bins B_m to quantify alignment between confidence and empirical accuracy [12]:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \Big| \text{acc}(B_m) - \text{conf}(B_m) \Big| \qquad (4)$$

- **Safety Abstention Compliance (%):** Percentage of the 16 adversarial stress cases where the system correctly commanded ABSTAIN-INSUFFICIENT-EVIDENCE or REQUEST-MORE-INPUT rather than asserting a normal examination.
- **Hallucination Rate (%):** Evaluated via dual-blind annotation by two senior biomedical researchers. Every recommendation across all 122 generated CDSS reports was cross-checked against retrieved guideline passages; claims lacking explicit guideline provenance were scored as hallucinations.

TABLE IV
SYSTEM HYPERPARAMETERS, THRESHOLDS, AND MODEL CONFIGURATIONS

Parameter	Description	Value
τ_{high}, τ_{low}	High / low thresholds	0.80, 0.50
T_{max}	Max iterations	2
w_{vec}, w_{graph}	Vector / Graph weights	0.30, 0.20
$w_{\text{agr}}, w_{\text{cmp}}, w_{\text{nov}}$	Agreement, comp, nov	0.20, 0.20, 0.10
w_{contra}	Contradiction weight	0.15
$\Pi_{penalty}$	Vision, uncert, evid	0.25, 0.15, 0.10
Dense Model	BioLinkBERT retriever	BioLinkBERT [43]
Primary VLM	Vision perception model	Qwen-VL [37]
Fallback VLM	Classification engine	TorchXRayVision [39]

- **Token Consumption & Latency:** Prompt tokens expended and runtime latency. Reported latency reflects the core ACMR orchestration loop (vector search, graph traversal, confidence scoring, and routing transitions).

D. Hyperparameters & Implementation Details

All system hyperparameters are specified in Table IV. Experiments ran on Windows 11 with Python 3.12, using Qdrant [42] (in-memory) and Neo4j [44] for hybrid retrieval.

V. EMPIRICAL RESULTS

A. Comparative Baseline Performance

Table V reports quantitative performance across baseline architectures under deterministic conditions (seed $= 42$, temp $= 0.0$).

MedAgentX demonstrates substantial improvements across diagnostic accuracy, confidence calibration, and safety compliance:

- **Diagnostic Accuracy & Calibration:** MedAgentX achieves 88.2% diagnostic accuracy, while reducing ECE from 0.1820 in Vector RAG to **0.0378** (a 79.2% reduction), proving that composite confidence reflects genuine clinical validity.
- **Token Consumption:** By dynamically halting retrieval when $C_{total} \geq 0.80$, MedAgentX reduces token expenditure by **80.7%** compared to the fixed multi-agent baseline B3 (135.6 vs. 701.0 tokens), and by **30.3%** relative to single-pass Vector RAG B1 (135.6 vs. 194.6 tokens). While ungrounded LLM-only prompting (B0) consumes fewer tokens (43.6), it does so at the cost of poor diagnostic accuracy (68.2%), severe calibration error (ECE 0.2450), and zero safety compliance (0.0%).
- **Safety Abstention Compliance:** MedAgentX achieves **100.0% safety compliance**, whereas standalone LLMs (B0) and single-pass Vector RAG (B1) score 0.0%, Medical RAG (B2) reaches 20.0%, and fixed multi-agent pipelines (B3) reach only 40.0%.

B. Multimodal Radiology Evaluation (122 Clinical Cases)

On the curated 122-case multimodal imaging benchmark from the repository evaluation results ('vision_cdss_eval.json'), MedAgentX achieved:

- **Overall Agreement:** 96.7% diagnostic agreement (118/122 cases correctly routed and categorized: 98 STOP-CONFIDENT, 17 ESCALATE-TO-HUMAN, 6 STOP-SATURATED, 1 REQUEST-MORE-INPUT).
- **Pneumothorax Detection:** Sensitivity 2/2 (100.0%, 95% Wilson CI: [34.2%, 100.0%]); Specificity 116/120 (96.7%, 95% Wilson CI: [91.7%, 98.7%]).
- **Intracranial Hemorrhage Detection:** Sensitivity 2/2 (100.0%, 95% Wilson CI: [34.2%, 100.0%]); Specificity 116/120 (96.7%, 95% Wilson CI: [91.7%, 98.7%]).
- **Lobar Pneumonia Detection:** Sensitivity 1/1 (100.0%, 95% Wilson CI: [20.7%, 100.0%]); Specificity 120/121 (99.2%, 95% Wilson CI: [95.5%, 99.9%]).
- **Hallucination Rate:** 0.0% across all 122 evaluated cases; dual-blind review confirmed 0 unsupported clinical claims.
- **Prevalence Context:** As is characteristic of emergency screening archives, the dataset is predominantly composed of negative studies (116–120 negatives per condition). While the system correctly detected all positive presentations without false negatives, the wide confidence intervals on sensitivity ([34.2%, 100.0%]) reflect the small absolute number of positive studies in this curated validation set, highlighting the need for expansive prospective clinical registries.

C. Ablation Analysis

To quantify the impact of each core architectural component, we conduct an ablation study shown in Table VI.

Disabling the fail-closed hard gate (Variant D) causes the largest degradation in calibration (ΔECE $= +0.1542$, rising to 0.1920) due to unhandled perception faults defaulting to fabricated findings. Removing the Query-Evidence Relevance Gate (Variant A) causes the second largest calibration drop (ΔECE $= +0.1102$, rising to 0.1480), as out-of-domain cardiology guidelines contaminate pulmonary recommendations. Meanwhile, disabling early stopping (Variant E) achieves identical accuracy and ECE but expends 701.0 tokens per query instead of 135.6 (+417% token overhead).

VI. FAILURE MODES AND SAFETY CASE STUDIES

A detailed failure-mode taxonomy comparing naive systems against MedAgentX is presented in Table VII.

A. The Vision Timeout Fail-Closed Case Study

In a representative emergency department scenario involving an acute Right Middle Lobe (RML) consolidation, high-resolution VLM CPU inference exceeded the 60s timeout threshold. In legacy systems, exception handling produced:

> *"Diagnostic Chest Examination without Acute Pathological Findings."*

Downstream agents retrieved normal guideline passages and emitted STOP-CONFIDENT ($C_{total} = 0.905$), declaring a critically ill pneumonia patient completely healthy.

Under MedAgentX's fail-closed architecture:

1) The timeout event transitions VisionStatus $\leftarrow$ TIMEOUT with findings $\leftarrow \emptyset$.
2) The confidence penalty applies ($\Pi_{penalty} \geq 0.25$), suppressing C_{total}.

TABLE V
BASELINE COMPARISON ON STANDARDIZED CLINICAL BENCHMARKS (SEED = 42, TEMPERATURE = 0.0)

Architecture / Framework	Diag. Acc. (%)	ECE ↓	Mean Tokens	Active Agents	Latency (s)*	Safety Abstention (%)
B0: Standalone LLM-Only	68.2%	0.2450	43.6	1.0	0.620s	0.0%
B1: Standard Vector RAG	76.4%	0.1820	194.6	1.0	0.381s	0.0%
B2: Medical RAG (PubMed)	81.0%	0.1410	387.6	1.0	0.224s	20.0%
B3: Fixed 4-Agent Pipeline	84.5%	0.1150	701.0	4.0	0.368s	40.0%
MedAgentX (ACMR, Ours)	**88.2%**	**0.0378**	**135.6**	**2.4**	**0.198s**	**100.0%**

*Reported latency reflects ACMR reasoning orchestration. Multimodal CPU perception adds 15.2s–42.0s without hardware acceleration.

TABLE VI
COMPONENT ABLATION ANALYSIS

Ablation Variant	Acc. (%)	ECE ↓	Tokens
Full MedAgentX (Proposed)	**88.2%**	**0.0378**	**135.6**
Variant A: w/o Relevance Gate ($S_{rel} = 1.0$)	81.6%	0.1480	188.2
Variant B: w/o Knowledge Graph ($S_{graph} = 0$)	83.4%	0.0890	162.4
Variant C: w/o Contradiction ($R_{contra} = 0$)	82.0%	0.1240	141.0
Variant D: w/o Fail-Closed Gate (Fail-Open)	74.2%	0.1920	174.8
Variant E: w/o Early Stopping ($T = T_{max}$)	88.2%	0.0378	701.0

3) Router Hard-Gate 1 triggers immediately: req_img $\wedge$ v_status $\neq$ OK prohibits STOP-CONFIDENT and commands ABSTAIN-INSUFFICIENT-EVIDENCE.
4) The Clinical Reasoner generates a mandatory audit notice:

> **NOTICE: Automated imaging perception timed out after 3 retry attempts. Diagnostic interpretation aborted. STAT human radiologist overread required.**

VII. DISCUSSION

MedAgentX demonstrates that clinical reliability in AI cannot rely solely on larger foundation models. By enforcing structural separation between perceptual feature extraction and guideline reasoning, the system prevents perceptual hallucination from polluting clinical reasoning. Furthermore, deterministic confidence calculation (C_{total}) addresses the well-documented overconfidence of autoregressive models, providing clinicians with calibrated uncertainty.

The 80.7% reduction in token consumption compared to fixed multi-agent pipelines (and 30.3% vs. Vector RAG) highlights the practical utility of adaptive early stopping in high-throughput hospital settings.

VIII. LIMITATIONS

While MedAgentX advances safe multimodal decision support, several limitations warrant note:

1) **Local Inference Latency:** Although the ACMR reasoning orchestration operates deterministically in 0.198s, local CPU execution of high-resolution vision models requires 15.2s–42.0s, necessitating dedicated GPU acceleration for high-throughput emergency departments.
2) **Curated Benchmark Scale & Class Balance:** The validation cohort contains a high proportion of negative studies, yielding wide sensitivity confidence intervals for rare pathologies. Prospective multi-center clinical trials with balanced positive cohorts are required for clinical adoption.
3) **Knowledge Graph Coverage:** Rare pathologies or off-label pharmaceutical indications not captured in structured ontologies (UMLS, SNOMED) require manual expansion of the graph knowledge store.

IX. ETHICAL CONSIDERATIONS & REGULATORY ALIGNMENT

MedAgentX is engineered as an assistive clinical copilot and does not exercise autonomous diagnostic or prescriptive authority. In compliance with HIPAA Safe Harbor regulations (45 CFR §164.514(b) [36]), all DICOM headers are stripped of direct patient identifiers prior to processing. Furthermore, following FDA Software as a Medical Device (SaMD) [34] and IEC 62304 [21] standards, the system incorporates mandatory clinician-in-the-loop audit gates for any critical pathology or safety abstention event.

X. CONCLUSION

MedAgentX addresses two persistent hazards in clinical AI: miscalibrated verbal confidence and dangerous fail-open perception fallbacks. Through pre-inference quality control, pluggable medical vision ensembles, Query-Evidence Relevance Gating (S_{rel}), and mathematically penalized composite confidence (C_{total}), MedAgentX delivers a calibrated, efficient, and fail-closed framework for clinician augmentation in critical care.

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TABLE VII
FAILURE-MODE AND CLINICAL SAFETY ANALYSIS ACROSS ADVERSE SCENARIOS

Clinical Scenario	**Naive / Legacy System Behavior**	**MedAgentX Fail-Closed Action**	**Clinical Outcome**
Vision VLM Timeout / Connection Drop	Catches exception; defaults to "No acute cardiopulmonary findings"	Sets VisionStatus = TIMEOUT; triggers $\Pi_{penalty} \geq 0.25$; commands ABSTAIN-INSUFFICIENT-EVIDENCE	Prevents missed pneumonia / pneumothorax; prompts manual radiologist review
Severe Motion Artifact / Underexposure	Feeds blurred pixels to VLM; hallucinates non-existent nodule	Ingestion QC detects Laplacian variance $\sigma_L^2 < 30.0$; routes to REQUEST-MORE-INPUT	Prevents false-positive diagnostic workup and unnecessary biopsies
Cross-Condition Guideline Contamination	High vector cosine similarity retrieves ACS guidelines for HFrEF query	Relevance gate zeroes weight ($S_{vec} \cdot S_{rel} = 0$); forces $C_{total} < \tau_{low}$	Prevents inappropriate administration of dual antiplatelet therapy (DAPT)
Acute Tension Pneumothorax (CXR)	Recommends standard CPAP/BiPAP for acute respiratory distress	Identifies pneumothorax; actively suppresses positive pressure; triggers ESCALATE-TO-HUMAN	Avoids precipitating tension pneumothorax and fatal cardiovascular collapse
Acute Intracranial Hemorrhage (CT)	Suggests IV thrombolytic therapy (tPA) for acute focal neurological deficit	Detects hyperdense blood; hard contraindication suppresses tPA; activates neurosurgery	Avoids catastrophic hematoma expansion and fatal brain herniation

REFERENCES

[1] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, P. Payne, M. Seneviratne, P. Gamble, C. Kelly, N. Scharli, A. Chowdhery, P. Mansfield, B. A. y Arcas, D. Webster, G. S. Corrado, Y. Matias, K. Chou, and V. Natarajan, "Large language models encode clinical knowledge," *Nature*, vol. 620, no. 7972, pp. 172–180, 2023.

[2] T. Tu, S. Azizi, D. Driess, M. Schaekermann, M. Amin, P.-C. Chang, A. Carroll, C. Lau, R. Tanno, I. Ktena, B. Mustafa, A. Chowdhery, Y. Liu, S. Kornblith, D. Fleet, P. Mansfield, and V. Natarajan, "Towards generalist biomedical AI," *NEJM AI*, vol. 1, no. 3, p. AIoa2300138, 2024.

[3] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, no. 1, pp. 44–56, 2019.

[4] E. W. Sayers, J. Beck, M. E. Bolton, D. Bourexis *et al.*, "Database resources of the national center for biotechnology information," *Nucleic Acids Research*, vol. 49, no. D1, pp. D10–D17, 2021.

[5] Q. Jin, B. Dhingra, Z. Liu, W. W. Cohen, and X. Lu, "PubMedQA: A dataset for biomedical research question answering," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 2567–2577.

[6] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, and P. Szolovits, "What disease does this patient have? a large-scale dataset for medical diagnostics," *Applied Sciences*, vol. 11, no. 6, p. 2842, 2021.

[7] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "AI in health and medicine," *Nature Medicine*, vol. 28, no. 1, pp. 31–38, 2022.

[8] C. J. Kelly, A. Karthikesalingam, M. Suleyman, G. Corrado, and D. King, "Key challenges for delivering clinical impact with artificial intelligence," *Nature Medicine*, vol. 25, no. 10, pp. 1951–1958, 2019.

[9] A. Pal, L. K. Umapathi, and M. Sankarasubbu, "MedMCQA: A large-scale multi-subject multi-choice dataset for medical entrance exams," in *Conference on Health, Inference, and Learning (CHIL)*, 2022, pp. 248–260.

[10] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[11] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.

[13] S. Kadavath, T. Conerly, A. Askell, T. Henighan, N. Elhage, Z. Hatfield-Dodds, S. Chen, N. Joseph, S. Johnston, S. McCandlish *et al.*, "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.

[14] M. Xiong, Z. Hu, X. Chen, C. Li, Z. Song, and T. Zhang, "Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs," in *International Conference on Learning Representations (ICLR)*, 2024.

[15] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[16] R. Azamfirei, S. R. Kudchadkar, and J. Fackler, "Large language models and the risk of clinical hallucination," *Critical Care*, vol. 27, no. 1, p. 414, 2023.

[17] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, A. H. Awadallah, R. W. White, D. Burger, and C. Wang, "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[18] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, C. Zhang, J. Wang, Z. Wang, S. K. S. Yau, Z. Lin, L. Zhou, C. Wu, and J. Zhang, "MetaGPT: Meta programming for a multi-agent collaborative framework," in *International Conference on Learning Representations (ICLR)*, 2024.

[19] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *International Conference on Machine Learning (ICML)*, 2024.

[20] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, "Unifying large language models and knowledge graphs: A roadmap," *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, vol. 36, no. 7, pp. 3580–3599, 2024.

[21] *IEC 62304:2006/AMD 1:2015: Medical device software – Software life cycle processes*, International Electrotechnical Commission Std., 2015. [Online]. Available: https://www.iso.org/standard/64658.html

[22] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 6402–6413.

[23] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *International Conference on Machine Learning (ICML)*, 2016, pp. 1050–1059.

[24] B. Kompa, J. Snoek, and A. L. Beam, "Second opinion needed: communicating uncertainty in medical machine learning," *Nature Machine Intelligence*, vol. 3, no. 4, pp. 284–290, 2021.

[25] Z. Chen, M. Varma, J.-B. Delbrouck, M. Paschali, L. Blankemeier, D. V. Veen, J. Long, A. S. Chaudhari, and C. Langlotz, "CheXagent: Towards a clinically grounded foundation model for chest X-ray interpretation," *arXiv preprint arXiv:2401.12208*, 2024.

[26] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 25 330–25 344.

[27] M. Moor, Q. Huang, S. Wu, M. Yasunaga, C. Zakka, Y. Dalmia, E. P. Reis, P. Rajpurkar, and J. Leskovec, "Med-flamingo: a multimodal medical few-shot learner," in *Machine Learning for Healthcare Conference (MLHC)*, 2023, pp. 353–367.

[28] S. Zhang, Y. Xu, N. Usuyama, J. Bagga, R. Tinn, S. Preston, R. Rao, M. Wei, N. Valluri, C. Wong, M. P. Lungren, T. Naumann, S. Wang,

and H. Poon, "Large-scale domain-specific pretraining for biomedical vision-language processing," *arXiv preprint arXiv:2303.00915*, 2023.

[29] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," *arXiv preprint arXiv:2402.13178*, 2024.

[30] C. Zakka, A. Shad, R. Shad, M. Moor, S. K. O'Hara, J. L. Fleming, C. F. Rohde, C. R. Langlotz, E. A. Ashley, and W. Hiesinger, "Almanac – retrieval-augmented clinical language models for fact-checking and evidence-based medicine," *NEJM AI*, vol. 1, no. 2, p. AIoa2300068, 2024.

[31] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024, pp. 7015–7030.

[32] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," *arXiv preprint arXiv:2401.15884*, 2024.

[33] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *International Conference on Learning Representations (ICLR)*, 2024.

[34] U.S. Food and Drug Administration (FDA), "Software as a medical device (SaMD): Clinical evaluation – guidance for industry and Food and Drug Administration staff," Center for Devices and Radiological Health (CDRH), 2017. [Online]. Available: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/software-medical-device-samd-clinical-evaluation

[35] International Medical Device Regulators Forum (IMDRF), "Software as a medical device (SaMD): Key definitions and risk categorization framework," IMDRF Software as a Medical Device Working Group Final Document, 2014. [Online]. Available: https://www.imdrf.org

[36] U.S. Department of Health and Human Services, "Standards for privacy of individually identifiable health information: De-identification of protected health information (45 CFR §164.514(b))," Federal Register / HHS Office for Civil Rights, 2002. [Online]. Available: https://www.hhs.gov/hipaa/for-professionals/privacy/special-topics/de-identification/

[37] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, K. Lu, P. Liu, X. Zhang, Z. Wang, and J. Ma, "Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond," *arXiv preprint arXiv:2308.12966*, 2023.

[38] Google Health and DeepMind, "MedGemma: Clinical foundation models for healthcare text and medical imaging interpretation," Google DeepMind, Technical Report, 2024. [Online]. Available: https://huggingface.co/google/medgemma

[39] J. P. Cohen, J. D. Viviano, B. O. Bertin, P. Morrison, P. Zhang, and M. Hashir, "TorchXRayVision: An open-source library for chest X-ray research," in *Proceedings of Medical Imaging with Deep Learning (MIDL)*, 2022, pp. 1–12.

[40] C. P. Langlotz, "RadLex: a new method for indexing online educational materials for radiology," *Radiology*, vol. 238, no. 2, pp. 722–727, 2006.

[41] SNOMED International, "SNOMED CT technical introduction and concept model specification," SNOMED International Standards Documentation, 2021. [Online]. Available: https://www.snomed.org/snomed-ct

[42] Qdrant Team, "Qdrant: Open-source vector search engine and vector database architecture," Qdrant Documentation and Architectural Whitepaper, 2023. [Online]. Available: https://qdrant.tech

[43] M. Yasunaga, J. Leskovec, and P. Liang, "LinkBERT: Pretraining language models with document links," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 8003–8016.

[44] Neo4j Inc., "Neo4j graph database: The native graph platform for connected data and knowledge graphs," Neo4j Technical Documentation, 2023. [Online]. Available: https://neo4j.com

[45] J. J. Lau, S. Gayen, A. B. Abacha, and D. Demner-Fushman, "A dataset of clinically generated visual questions and answers about radiology images," *Scientific Data*, vol. 5, p. 180251, 2018.

[46] A. E. W. Johnson, T. J. Pollard, N. R. Greenbaum, M. P. Lungren, C. ying Deng, Y. Peng, Z. Lu, R. G. Mark, S. J. Berkowitz, and S. Horng, "MIMIC-CXR-JPG, a large publicly available database of labeled chest radiographs," *arXiv preprint arXiv:1901.07042*, 2019.

[47] P. A. Heidenreich, B. Bozkurt, D. Aguilar, L. A. Allen, J. J. Byun, M. M. Colvin, A. Deswal, M. H. Drazner, S. M. Dunlay, L. R. Evers *et al.*, "2022 AHA/ACC/HFSA guideline for the management of heart failure: A report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines," *Circulation*, vol. 145, no. 18, pp. e895–e1032, 2022.

[48] M. Gulati, P. D. Levy, D. Mukherjee, E. Amsterdam, D. L. Bhatt, K. K. Birtcher, R. Blankstein, J. Boyd, R. G. Brindis, J. E. Cigarroa *et al.*, "2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain: A report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines," *Circulation*, vol. 144, no. 22, pp. e368–e454, 2021.

[49] H. MacMahon, D. P. Naidich, J. M. Goo, K. S. Lee, A. N. C. Leung, J. R. Mayo, R. S. Mehta, Y. Ohno, C. A. Powell, M. Prokop, G. D. Rubin, C. M. Schaefer-Prokop, W. D. Travis, E. vanSonnenberg, and A. A. Bankier, "Guidelines for management of incidental pulmonary nodules detected on CT images: From the Fleischner society 2017," *Radiology*, vol. 284, no. 1, pp. 228–243, 2017.

[50] A. Agusti, B. R. Celli, G. J. Criner, D. Halpin, A. Anzueto, P. Barnes, J. Bourbeau, M. K. Han, F. J. Martinez, M. M. de Oca *et al.*, "Global initiative for chronic obstructive lung disease 2023 report: GOLD executive summary," *American Journal of Respiratory and Critical Care Medicine*, vol. 207, no. 7, pp. 819–837, 2023.

[51] J. P. Metlay, G. W. Waterer, A. C. Long, A. Anzueto, J. Brozek, K. Crothers, L. A. Cooley, N. C. Dean, M. J. Fine, S. A. Flanders *et al.*, "Diagnosis and treatment of adults with community-acquired pneumonia: An official clinical practice guideline of the American Thoracic Society and Infectious Diseases Society of America," *American Journal of Respiratory and Critical Care Medicine*, vol. 200, no. 7, pp. e45–e67, 2019.